

Remote art therapy in collaborative virtual environment: a pilot study on feasibility and usability

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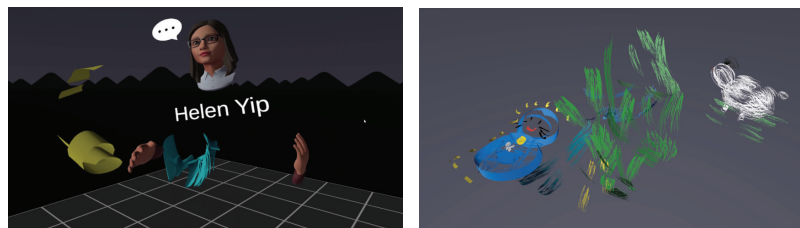


Figure 1: The client's view of the therapist in the CVE (left) and the artwork co-created by the client (C1) and the therapist (right).

ABSTRACT

The uniqueness of the collaborative virtual environment (CVE) makes it a perfect medium for delivering art therapy sessions remotely during the global pandemic. This pilot study investigates the feasibility and usability of delivering remote art therapy in a customised CVE. Four young adults as clients and two registered expressive arts therapists participated in the study. The quantitative and qualitative results suggested that the CVE-enabled approach was feasible and the system's usability was high. The approach was well received by both the clients and the therapists. Possible improvements to the CVE were identified and will be addressed in our future work.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI;

1 INTRODUCTION

Art therapy is a term that umbrellas the use of various art forms (e.g., painting, music, dance, drama, etc.) to improve one's psychological well-being by employing psychoeducational, humanistic, and systemic approaches [7, 10]. The practice of art therapy is mainly based on the idea that creative processing can facilitate the non-verbal communication of feelings and thoughts [7]. As the global pandemic stays, remote practices of psychotherapy assisted by information and communication technologies are getting increasingly popular [9]. However, the practice of remote art therapy faces the challenge of effectively translating and maintaining the unique triangular therapeutic relationship among the therapist, the client, and the artworks in the digital world [2]. Collaborative virtual environments (CVE) seem to be a perfect medium for delivering painting and sculpturing art therapy remotely. CVE has been appraised for enabling a strong sense of social presence among interaction partners - the subjective experience of being present with a "real" person and "having access to his or her thoughts and emotions" [8] (p. 1). Moreover, it

allows the therapist and the client to work on the three-dimensional artworks collaboratively and simultaneously, bringing the possibility of translating and maintaining the triangular therapeutic relationship in the virtual world. Although there is not much work on using CVE for remote art therapy, a few pilot studies have been conducted to investigate the feasibility of using off-the-shelf virtual reality (VR) applications to enable art therapy. For example, Haeyen et al. invited five art therapists and two psychomotor therapists to experiment with Tilt Brush¹ in their clinical practice [3]. Kaimal et al. conducted a similar study to collect qualitative feedback from participants with more diverse backgrounds outside the clinical setting regarding the use of Tilt Brush¹ and Kodon² for art making [5]. In both studies, the VR applications were considered a promising medium for enabling art therapy and were generally welcomed by the participants. However, because these off-the-shelf VR applications were designed to provide single-user experiences, therapists could only observe the art creation process on computer monitors without actual participation.

This pilot study investigates the feasibility and usability of delivering remote art therapy in a highly customised CVE by collecting quantitative and qualitative evidence. Our results suggest that the approach was feasible and well-received by both the clients and the therapists. Possible improvements to the CVE were identified and will be addressed in our future work.

2 METHOD

2.1 Collaborative Virtual Environment

We designed and implemented a highly-customised CVE for this pilot study (see Figure 1). Besides real-time collaborative art creations, the CVE also supports real-time voice chat and browser-based avatar customisation. To avoid the technical complexity and to induce greater co-presence as suggested by [4], instead of supporting full-body avatars, only half-body avatars with head and hand tracking were presented in the CVE. The typical client-server software architecture was employed, and the CVE was developed using Unity³. Photon for Unity (PUN)⁴ network framework was used to ease the network programming and enable a reliable low-latency voice chat. To synchronise the artworks on multiple instances of the software, the art creation process was decomposed into a series of pre-defined remote procedure calls (RPCs).

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¹<https://www.tiltbrush.com/>

²<https://www.kodon.xyz/>

³<https://unity.com/>

⁴<https://www.photonengine.com/pun>

Table 1: Age and gender of the clients

ID	C1	C2	C3	C4
Gender	Female	Male	Male	Male
Age (years)	20	24	22	19

Table 2: Results of the SUS scores

ID	C1	C2	C3	C4	T1	T2
SUS	87.5	77.5	72.5	75	77.5	82.5

2.2 Participants

Four young adults were referred by a non-governmental organisation (NGO) to participate in this pilot study as clients (C1, C2, C3 and C4). The age and gender of the clients are shown in Table 1. Except for C3, who had previous experience playing VR games, other clients had no experience before the study. Two registered expressive arts therapists (T1 and T2) also participated in the study. The two therapists had previous experiences in delivering remote arts therapy using video conferencing tools during the global pandemic but had no VR experience. All participants had a normal or corrected-to-normal vision. The study was approved by the Institutional Review Board (application no.: HSEARS20220517004).

2.3 Procedure

Each participant, including the four clients and two therapists, had a 30-minute one-on-one session with the fourth author of this paper, a registered expressive arts therapist who received prior training on using the CVE. During the sessions, the clients were told to draw a comfortable and cosy space in VR, and the therapists were asked to evaluate the system as a medium for delivering remote art therapy. The instruments for measuring system usability and cybersickness symptoms were administered after the session, followed by a semi-structured interview focusing on possible improvements to the CVE.

2.4 Measures

The System Usability Scale (SUS) [1] was used to measure the usability of our custom-designed CVE. The scale is the most widely used instrument to conduct subjective assessments of usability. The Simulator Sickness Questionnaire (SSQ) [6] was used to measure the cybersickness symptoms experienced by the participants.

3 RESULTS AND DISCUSSION

3.1 System Usability and Cybersickness

The results of the SUS and SSQ scores are summarised in Table 2 and Figure 2 respectively. The SUS scores suggested that the usability of the CVE was good. Generally speaking, using the CVE for remote art therapy also did not introduce more cybersickness symptoms, as reflected by the SSQ scores, except for C2, who said that he had a long history of being sensitive to motion sickness.

3.2 Qualitative Feedback

C1, C3, and C4 found the collaborative art creation process “enjoyable and interesting”. C2 said that although the process was interesting, he felt discomfort after around 10 minutes in the CVE. C1 and C4 expressed their interest in having more sessions with the therapist (i.e., the fourth author of the paper). C3 said he felt “the session was a little too short” since he did not get “familiar with all the features [of the CVE]”. C3 suggested having more brushes and C4 wanted to have a feature (e.g., a virtual mirror) that he could see his self-avatar better. Both T1 and T2 agreed that the CVE was a promising medium for delivering remote arts therapy. T1 said that she liked “the idea of having more kinaesthetic movements during art creation”. T2 said that “creating artworks in three-dimensional spaces was amazing” and she would like to try it out with her clients.

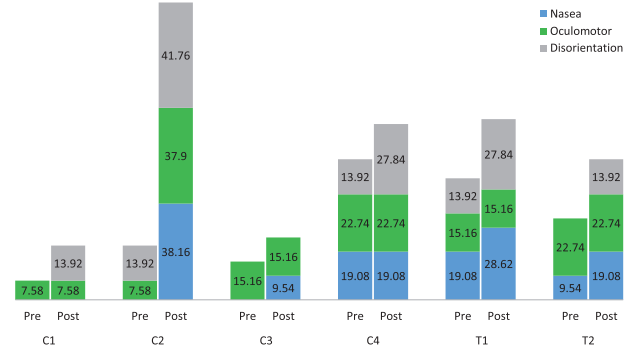


Figure 2: Results of the SSQ scores

T2 also mentioned the preference for having more modalities (e.g., music) for art creations. Both T1 and T2 believed the software system was easy to use and their clients should have no issues participating in remote art therapy using the CVE.

4 CONCLUSION AND FUTURE WORK

Motivated by the need for remote art therapy during the global pandemic and the unique features of CVE, we conducted this pilot study to investigate the feasibility and usability of a highly customised CVE for delivering remote arts therapy. The quantitative and qualitative results suggested that the CVE-enabled approach was feasible and the system’s usability was high. The approach was well received by both the clients and the therapists. Possible improvements to the CVE were identified through semi-structured interviews. We plan to further investigate the approach’s effectiveness in the clinical setting after identified improvements have been made.

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